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Recalculation Conditions

Calculated fields can be recalculated when a user is viewing or editing a record. To initiate an immediate recalculation, a user must have update permissions to the record. When initiated from either mode, only marked content is recalculated. Content changes may result in outstanding calculations in a related level or application.

The recalculation can be initiated when content is changed, or for content that has a status of marked for recalculation. All calculated fields are recalculated immediately within the current content. All related content affected by the change is marked and queued for recalculations in an asynchronous job. When save or apply updates calculated fields, and there are no other user changes, notifications are not sent. The recalculate action recalculates all active calculated fields for the current record regardless of any other conditions or configurations.

When marked content is calculated asynchronously, only the fields associated with the executed job are calculated. Notifications are never sent. Notifications are only sent when a user saves a record.

Calculated fields are only recalculated based on changes made directly in a data feed, data import, web API, or scheduled recalculation jobs.

On this page

More information about recalculations in calculated fields

- In addition to scheduled recalculations, field recalculations are performed for a record each time a user clicks Save or Apply for the record.
- Search does not trigger a recalculation of field values.
- Scheduled recalculations are written directly to the database and are not interpreted by the application as true “record save” events and are not captured in the History Log field.
- Scheduled recalculations do not trigger notifications.
- Field-value changes stemming from a scheduled recalculation are not reflected in the audit information displayed alongside a field.
- Each time that you create or edit a calculated field, the system searches for NOW and TODAY in all of the application or formulas of the sub-form. If the system can no longer locate either of these functions, any previously configured recalculation schedule are automatically disabled for the application or sub-form.
- Fields with the As Needed option selected for recalculations are only recalculated if the value will be changed.
- In multi-level applications, recalculation schedules are level-specific.

The following table describes other conditions that may also trigger a recalculation condition.

Trigger	Description
Full Module Calculation	All fields for all content are queued for recalculation but are not marked.
Scheduled Recalculations	Administrator creates a Schedule to run an asynchronous job for recalculation at a specific time. Only the fields that contain NOW() or TODAY() functions in their formulas are recalculated. If the record contains such fields, and any of them recalculate to a different value, then any fields set to Always Recalculate in the record will also be recalculated.

Trigger	Description
Related Content	When a user, job engine, data feed, or Web API makes a change to a field that affects changes in a related module, the following occurs: • An asynchronous job is scheduled to recalculate all affected content 1 level away of the related module, for example related record in current application or cross-referenced application in related record. • All affected content in the related module (1 level away) is marked for recalculation.

Recalculation and error handling rules

The following table describes recalculation and error handling rules.

Rule	Description
Recalculation	Determines when a field is recalculated As Needed or Always. As Needed: Formulas are recalculated when a dependent field in the formula changes. Always: Formulas are recalculated every time content is saved even though a field is not referenced in the formula. Formulas that contain NOW() and TODAY() functions, or user first name, last name, and middle name (Editor) parameters are recalculated regardless of content change.
Error Handling	Determines what happens when a calculation error occurs. This rule has the following options: Display Error: The word Error is displayed as a link when a calculation error occurs. Users with the appropriate access privileges can click the link to open the Calculation Error page where the error is explained. Use No Value: An empty value is saved in the field when a calculation error occurs. Use Specific: A specific value is saved in the field when a calculation error occurs.

Recalculations in edit mode

When in edit mode, the Recalculate button is not available. The recalculation is initiated from the Save button.

Example: Cross-referenced field updated

The following table describes the example.

Scenario	Calculated field is [Total Risk] in Application A. [Risk] is a cross-referenced field. [Controls] is a level in the cross-reference multi-level application and [Severity Rating] is a field in the Controls data level. SUM(REF([Risk], [Severity Rating], [Controls]))

Action 1	User drills into [Severity Rating] in Application B. [Severity] = 12. User changes value of [Risk] to 11 and clicks Save. Content of [Total Risk] is 'marked' for recalculation.
Action 2	User with Read and Update permissions returns to Application A in Edit mode and clicks Save.
Action 3	User saves record in Application B.
Results	[Total Risk] is recalculated immediately, and the updated value is displayed. [Total Risk]=23

Example: Calculated field updated by data feed

The following table describes the example.

Scenario	Application A has 3 fields Risk, Criticality, and Severity. Rating is a related record in Application B. [Total Risk] is dependent on the value of [Criticality].total SUM(REF([Risk], [Criticality])) [Severity] is dependent on [Rating]. IF([Rating]=10, VALUEOF ([Severity],"High"),VALUEOF ([Severity], "Low"))
Action 1	User changes the value of Criticality in Application B and clicks Save.
Results	Related content in Application A is 'marked'.
Action 2	Data feed updates [Rating] in Application A to a value of 10. Severity is calculated upon content save initiated by the data feed.
Results	[Total Risk] is not recalculated. [Rating] is updated during the data feed. [Severity] is changed to High.
Action 3	User view records and clicks Recalculate.
Results	[Total Risk] is recalculated immediately.

Recalculations in view mode

When in View mode, a message is displayed stating that the content may not be current. The Recalculate button is available.

Example: Cross-Referenced field updated

The following table describes the example.

Scenario	Calculated field is [Total Risk] in Application A. [Risk] is a cross-referenced field. [Controls] is a level in the cross-reference multi-level application and [Severity Rating] is a field in the Controls data level. SUM(REF([Risk], [Severity Rating], [Controls]))

Action 1	User edits [Severity Rating] in Application B. [Severity] = 12. User changes value of [Risk] to 11 and clicks Save. Content of [Total Risk] is ‘marked’ for recalculation.
Action 2	User with Read and Update permissions returns to Application A in View mode and clicks Recalculate.
Action 3	User saves record in Application B.
Results	[Total Risk] is recalculated immediately, and the updated value is displayed. [Total Risk]=23

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